Learning Objective 3.9: Calculate null-steering weights, implement pattern nulls on the ADALM-PHASER, and compare manual null steering with adaptive beamforming.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **From weights to GUI settings.** A null-steering calculation for a 4-element subarray returns the complex weights

$$w_1 = 0.62 - j0.13, \quad w_2 = 0.88 + j0.05, \quad w_3 = 0.88 - j0.05, \quad w_4 = 0.62 + j0.13$$

The Phaser GUI takes element amplitude as a percentage of the largest weight and element phase as an offset in degrees.

(a) Compute the four magnitudes and convert them to the percentages you would type into Element Gains.

gains =

(b) Compute the four phase offsets in degrees.

phases =

(c) A student enters the four gain percentages but forgets to enter the phase offsets. Describe the pattern they will measure and say whether a notch appears.

(d) All four weights are multiplied by e^{j40° before being entered. How does the measured pattern change, and why?

2. **Reading a null-steered sweep.** A student sweeps the uniform 8-element array, freezes the trace, enters null-steering weights for a null at $+22.5°$, and sweeps again. The two traces read as follows.

	Uniform (frozen)	**Null steered**
Peak level	0.0 dB (reference)	−1.8 dB
Level at $+22.5°$	−12.8 dBc	−21.6 dBc

(a) How much additional rejection did the null steering buy at $+22.5°$?

$\Delta =$ ☐

(b) Express the notch depth relative to the null-steered array's own peak, which is the number a jammer-to-signal calculation needs.

depth = ☐

(c) A second student reports a notch at −31 dBc on the same bench with the same weights. The sweep's noise floor sits about 22 dB below the uniform peak. Is the report credible? Explain.

(d) The notch appears centred at $+19.7°$ rather than $+22.5°$. Give the most likely cause.

3. **Null depth under coarse quantization.** The lab is repeated on a board whose phase shifters have only $B = 2$ bits.

   (a) Find the phase LSB.

   LSB = $\boxed{}$

   (b) Use the rule of thumb that quantization sidelobes sit near $-6B$ dB to estimate the deepest notch this board can hold.

   depth ≈ $\boxed{}$

   (c) Round the four phases from Question 1(b) to the nearest multiple of the 90° LSB. What does the resulting weight vector do?

   phases = $\boxed{}$

   (d) Explain why holding a deep null is more demanding of phase resolution than holding a beam peak.

4. **Two-channel MVDR by hand.** The PHASER's two digital channels are steered so that a target off boresight presents the real steering vector $s = [1, -1]^T$, while an interferer on boresight presents $u = [1, 1]^T$ with power $P = 9$ relative to a per-channel noise power of 1. The covariance seen by the beamformer is $R = \sigma^2 I + P\,uu^T$.

   (a) Build R.

   $R =$ $\boxed{}$

   (b) Compute the MVDR weights $w = R^{-1}s/(s^T R^{-1} s)$.

   $w =$ $\boxed{}$

   (c) Evaluate the beamformer's response to the target, $w^T s$, and to the interferer, $w^T u$.

   $w^T s,\ w^T u =$ $\boxed{}$

   (d) Which lab procedure did MVDR just reproduce, and what would w become if the interferer were switched off?

5. **Choosing an approach.** Each situation below uses the same 8-element PHASER.

   (a) A fixed ground emitter sits at a surveyed bearing of $+35°$ from the array face and does not move. The array must keep its beam on a target at $-10°$. Which approach would you use, and why?

   (b) An airborne jammer of unknown and changing bearing appears during a mission. Which approach would you use, and why?

   (c) A requirement calls for a 33 dB notch. Taking 21 dB as the depth achievable with the 7-bit ADAR1000 and roughly 6 dB of improvement per added bit, how many phase bits are needed?

   $B =$ $\boxed{}$

   (d) Two jammers now illuminate the array from different bearings. State what happens when MVDR is asked to handle both on this hardware.

Documentation: ____